

Exercise (CCA security for commitments). Consider an opening oracle $\mathcal{O}$ against a commitment scheme $(\text{Gen}, \text{Com}, \text{Open})$ that is defined in the following way

$$\mathcal{O}_{\text{pk}}(c) \left[\begin{array}{l} d_* \leftarrow_u \{d : \text{Open}_{\text{pk}}(c, d) \neq \perp\} \\ \text{return } d_* \end{array} \right. .$$

Let $\mathcal{L}$ denote the list of inputs submitted to $\mathcal{O}_{\text{pk}}$. Then the commitment scheme is (t, q, ε) -binding against opening oracle the advantage of any t -time adversary $\mathcal{A}$ in the following game

$$\mathcal{G} \left[\begin{array}{l} \text{pk} \leftarrow \text{Gen} \\ c, d_0, d_1 \leftarrow \mathcal{A}^{\mathcal{O}_{\text{pk}}(\cdot)}(\text{pk}) \\ \text{if } \text{Open}_{\text{pk}}(c, d_0) = \perp \text{ then } \text{return } 0 \\ \text{if } \text{Open}_{\text{pk}}(c, d_1) = \perp \text{ then } \text{return } 0 \\ \text{if } c \in \mathcal{L} \vee |\mathcal{L}| > q \text{ then } \text{return } 0 \\ \text{return } \text{Open}_{\text{pk}}(c, d_0) \neq \text{Open}_{\text{pk}}(c, d_1) \end{array} \right.$$

is bounded by ε , i.e., $\Pr[\mathcal{G}^{\mathcal{A}} = 1] \leq \varepsilon$. Assume that there exists a τ -time adversary $\mathcal{B}$ that has a large advantage $\text{Adv}(\mathcal{B})$ against the following game pairs

$$\begin{array}{ll} \mathcal{Q}_0 & \mathcal{Q}_1 \\ \left[\begin{array}{l} \text{pk} \leftarrow \text{Gen} \\ m_0, m_1 \leftarrow \mathcal{B}(\text{pk}) \\ \text{if } m_0 = m_1 \text{ then } \text{return } 0 \\ c, d \leftarrow \text{Com}_{\text{pk}}(m_0) \\ c_* \leftarrow \mathcal{B}(c) \\ d_* \leftarrow \mathcal{B}(d) \\ \text{if } c = c_* \text{ then } \text{return } 0 \\ \text{return } \text{Open}_{\text{pk}}(c_*, d_*) = m_0 \end{array} \right. & \left[\begin{array}{l} \text{pk} \leftarrow \text{Gen} \\ m_0, m_1 \leftarrow \mathcal{B}(\text{pk}) \\ \text{if } m_0 = m_1 \text{ then } \text{return } 0 \\ c, d \leftarrow \text{Com}_{\text{pk}}(m_1) \\ c_* \leftarrow \mathcal{B}(c) \\ d_* \leftarrow \mathcal{B}(d) \\ \text{if } c = c_* \text{ then } \text{return } 0 \\ \text{return } \text{Open}_{\text{pk}}(c_*, d_*) = m_1 \end{array} \right. \end{array}$$

where

$$\text{Adv}(\mathcal{B}) = \frac{1}{2} \cdot \Pr[\mathcal{Q}_0^{\mathcal{B}} = 1] + \frac{1}{2} \cdot \Pr[\mathcal{Q}_1^{\mathcal{B}} = 1] - \frac{1}{2} .$$

Show that then the commitment scheme cannot be $(t, 1, \varepsilon)$ -binding unless the original commitment is statistically binding. If it is not possible provide a counter example.

Solution. For simplicity analyse the case where each commitment c has only two valid openings d_0 and d_1 and both lead to different messages.